

On Poincaré series of valuations on subfields of the complex numbers

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Value semigroups and good semigroups.
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Standard/Main Example

- Let $\mathbb{K} \subset \mathbb{C}$ (e.g. $\mathbb{K} = \mathbb{Q}$), $F \in \mathbb{K}[[x, y]]$ (or $\mathcal{E}_{\mathbb{K}^2, 0}$) irreducible.
 $R = \mathbb{K}[[x, y]]/(F)$ a plane algebroid branch over $\mathbb{K}$. The integral closure $\bar{R}$ is the valuation ring R_ω of a discrete valuation ω , its residual field $\mathbb{K}' = R_\omega/m_\omega \subset \mathbb{C}$ is a finite algebraic extension of $\mathbb{K}$.
- F defines a reduced algebroid curve over the complex field $\mathbb{C}$, each branch C corresponds to an irreducible factor of F in $\mathbb{C}[[x, y]]$. The valuation $\nu = \nu_C$ defined by C on $\mathbb{K}[[x, y]]$ coincides with the valuation ω .

Definition

The *semigroup of values*: $S_C^{\mathbb{K}} := \{\nu(f) : f \in \mathbb{K}[[x, y]]\} \subset \mathbb{Z}_{\geq 0}$.

- For $v \in \mathbb{Z}_{\geq 0}$, let $J^{\mathbb{K}}(v) := \{f \in \mathbb{K}[[x, y]] : \nu(f) \geq v\}$.
- The quotients $J^{\mathbb{K}}(v)/J^{\mathbb{K}}(v+1)$ are $\mathbb{K}$ -vector subspaces of $\mathbb{C}$.

Definition

The $\mathbb{K}$ -Poincaré series

$$P_C(t)(= P_C^{\mathbb{K}}(t)) = \sum_{v=0}^{\infty} \dim_{\mathbb{K}} \left(J^{\mathbb{K}}(v)/J^{\mathbb{K}}(v+1) \right) \cdot t^v = \sum_{v=0}^{\infty} a_v t^v.$$

The $\mathbb{K}$ -semigroup Poincaré series of the valuation ν is $P_C^S(t) = \sum_{v \in S_C^{\mathbb{K}}} t^v$.

Remark. Recall that, if $\mathbb{K} = \mathbb{C}$, all the coefficients of the series $P_C(t)$ are 0 or 1. This is not the case, in general, for a proper subfield $\mathbb{K} \subset \mathbb{C}$. E.g. if $\mathbb{K} = \mathbb{R}$ the coefficients could be 0, 1, or 2.

General case/Settings

- $\mathbb{K} \subset \mathbb{C}$ (e.g. $\mathbb{K} = \mathbb{Q}$, $\mathbb{K} = \mathbb{R}$)
- $\mathbb{K}[[x, y]] \subset \mathbb{C}[[x, y]]$
- $(C, 0) \subset (\mathbb{C}^2, 0)$ a plane algebroid branch, i.e. defined by an irreducible parametrization

$$x = x(\tau) = \tau^m, \quad y = y(\tau) = \sum_{i \geq m} c_i \tau^i \in \mathbb{C}[[\tau]].$$

For $f \in \mathbb{C}[[x, y]]$:

$$f(x(\tau), y(\tau)) = a(f) \cdot \tau^{\nu(f)} + \dots (> \text{degree}), a(f) \neq 0.$$

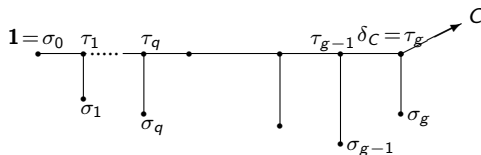
- C defines a valuation: $\nu = \nu_C : f \mapsto \nu(f) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$
- Consider its restriction to $\mathbb{K}[[x, y]]$, $\nu : \mathbb{K}[[x, y]] \rightarrow \mathbb{Z}_{\geq 0}$.

Complex branches. Dual graph

$\pi : (X, \mathcal{D} = \bigcup_{\sigma \in \Gamma} E_\sigma) \rightarrow (\mathbb{C}^2, 0)$ be an embedded resolution of C .

Let also denote the **dual graph** of π by Γ . For $\sigma \in \Gamma$:

- $\overset{\circ}{E}_\sigma$: smooth part of E_σ .
- $\gamma_\sigma = \{f_\sigma = 0\} \subset (\mathbb{C}^2, 0)$ a branch such that $\widetilde{\gamma}_\sigma$ is smooth and transversal to $\overset{\circ}{E}_\sigma$; $m_\sigma = \nu(f_\sigma)$.



- $m_{\sigma_0}, \dots, m_{\sigma_g}$ is the minimal set of generators of the semigroup of the complex branch C .

G -invariant resolution.

Let $G = \text{Gal}(\mathbb{C}/\mathbb{K})$ be the Galois group of $\mathbb{C}/\mathbb{K}$. The G -invariant resolution **essentially** is the resolution of $\bigcup_{g \in G} gC$.

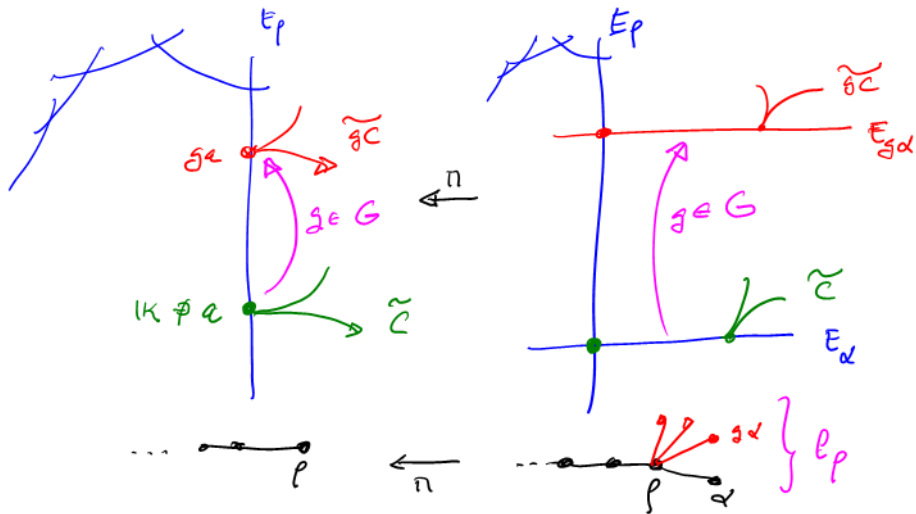
Let $\rho_1 = \rho \in \Gamma$ be the first divisor such that the strict transform of C (by the minimal modification $\pi_\rho : (X, D) \rightarrow (\mathbb{C}^2, 0)$ such that $E_\rho \subset D$) intersects E_ρ **at a point with coordinate $a \notin \mathbb{K}$** .

- 1) If a is algebraic over $\mathbb{K}$, then $\mathbb{K}_1 = \mathbb{K}(a)/\mathbb{K}$ is finite algebraic of degree $\ell_1 := \dim_{\mathbb{K}} \mathbb{K}_1 = [\mathbb{K}_1 : \mathbb{K}]$. Let $G_1 := \{g \in G : g(a) = a\} \subset G$. Notice that $\ell_1 = \#G/G_1$.

The different branches of the orbit of C by G intersects at ℓ_1 different points (the conjugates of a by G).

- 2) If a is transcendental over $\mathbb{K}$ we stop here. The branches from the G -orbit of C intersects at infinitely many different points.

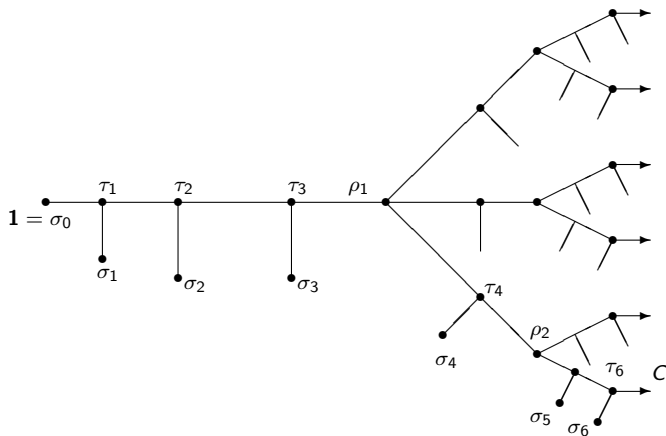
Vertices (divisors) as ρ_1 are the **splitting** points.



We have one of the next cases:

Case I)

There are finitely many branches in the G -orbit of C and they are resolved at a certain step. The dual graph Γ :



Case I)

The Galois group G acts on the graph Γ and the quotient $\check{\Gamma}$ by the action looks like:

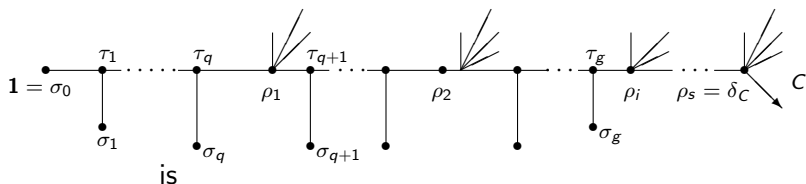


Figura: The resolution graph $\check{\Gamma}$ in Case I (with arbitrary g and s).

Case II)

The described process goes on without termination. This means that we get an infinite sequence of blow-ups. One can say that $s = \infty$ in this case.

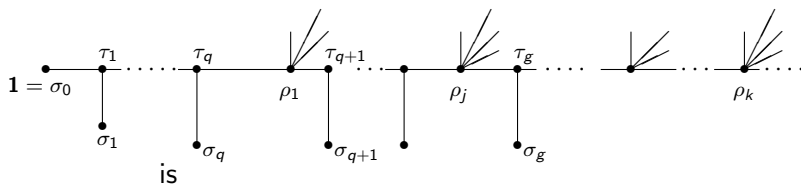


Figura: The resolution graph $\check{\Gamma}$ in Case II (with $s = \infty$).

Case III)

The process is terminated at a certain step. This means that at this step the branches from the G -orbit of C intersect each of the new-born components of the exceptional divisor at infinitely many points.

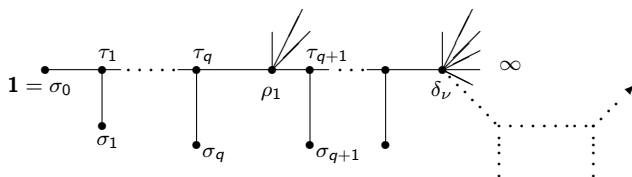


Figura: The resolution graph $\check{\Gamma}$ in Case III.

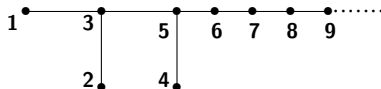
In this case, the valuation coincides with $n\nu$, where $\nu = \nu_{\rho_s}$ is the divisorial valuation defined by E_{ρ_s} and n is the intersection multiplicity $[E_{\rho_s}, \tilde{C}]$ of the strict transform $\tilde{C}$ of the curve C with the component E_{ρ_s} .

Example

Let $\mathbb{K} = \mathbb{Q}$ and the branch C given by the parametrization

$$x = \tau^4 \quad y = \alpha_0 \tau^6 + t^7 + \sum_{i \geq 1} \alpha_i \tau^{7+i}$$

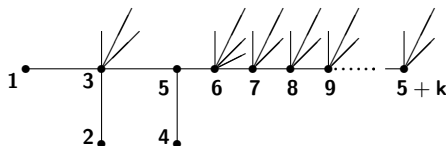
where $\alpha_0, \alpha_1, \dots \in \mathbb{C} \setminus \{0\}$. Let $\mathbb{K}' := \mathbb{Q}(\alpha_0, \alpha_1, \dots) \subset \mathbb{C}$.
The dual graph of C :



One can distinguish the following cases:

1. $\mathbb{K}'/\mathbb{Q}$ is finite algebraic. $\mathbb{K}' = \mathbb{Q}(\alpha_0, \dots, \alpha_k)$ for some k .

Assume $\alpha_i \notin \mathbb{Q}(\alpha_0, \dots, \alpha_{i-1})$ (algebraic, generic) for $0 \leq i \leq k$ then $s = k + 1$ and the splitting points are $\rho_1 = 3$, $\rho_i = 5 + (i - 1)$ for $i = 2, \dots, k + 1$. Thus we are in the Case I.



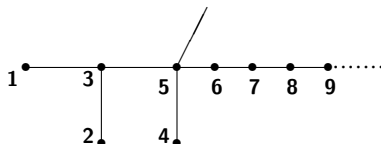
Pay attention that, if $\alpha_0 \in \mathbb{K}'$ is not generic, the picture can be somewhat different. For example: $y = \sqrt{2}\tau^6 + \tau^7$, ($gy = -\sqrt{2}\tau^6 + \tau^7$).

$$C : f(x, y) = (y^2 - 2x^3)^2 - 4\sqrt{2}yx^5 - x^7 = 0$$

$$gC : f(x, y) = (y^2 - 2x^3)^2 + 4\sqrt{2}yx^5 - x^7 = 0$$

$$C \cup gC : F = (y^2 - x^3)^4 - 32x^{10} - 2x^7(y^2 - x^3)^2 = 0.$$

In this case the only splitting point is $\rho = 5$.



2. $\mathbb{K}'$ is an infinite algebraic extension of $\mathbb{Q} = \mathbb{K}$. Then $s = \infty$ and we are in Case II. As in the previous case, in general, the splitting points are $3, 6, 7, \dots$
3. $\mathbb{K}'$ is a transcendental extension of $\mathbb{Q}$. In this case we are in Case III. For simplicity, let us assume that α_0 is transcendental over $\mathbb{Q}$. Then the process is interrupted in the divisor $\delta_\nu = 3$.

In order to get a function $g \in \mathcal{O}_{\mathbb{C}^2, 0}$ such that $\nu(g) = 2r + 1 \geq 7$ one needs that

$$g = (y^2 - \alpha_0^2 x^3) + \text{higher order terms}.$$

But in this case the equation can not be rational.

So, the semigroup $S_\nu^{\mathbb{Q}}$ is $\langle 4, 6 \rangle \subset 2\mathbb{Z}$ and the valuation ν is 2 times the divisorial valuation defined by the component E_3 .

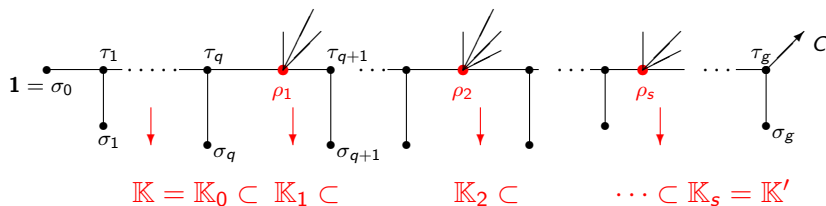
Isotropy subgroups and subfields.

Let G_σ be the isotropy subgroup of the component E_σ , i. e., the isotropy subgroup of the vertex σ in Γ .

- For all the vertices $\sigma \in \Gamma$ such that $\sigma_0 = \mathbf{1} \leq \sigma \leq \rho_1$ one has:
 $G_\sigma = G = G_0, \mathbb{K}_\sigma = \mathbb{K} = \mathbb{K}_0.$
- For all vertices $\sigma \in \Gamma, \rho_j < \sigma \leq \rho_{j+1}$: $G_\sigma = G_j, \mathbb{K}_\sigma = \mathbb{K}_j.$
- In Case I, for all the vertices $\sigma \in \Gamma, \rho_s < \sigma \leq \delta_C$ one has $G_\sigma = G_s,$
 $\mathbb{K}_\sigma = \mathbb{K}_s.$

One has $G = G_0 \supset G_1 \supset \dots \supset G_s, \mathbb{K} = \mathbb{K}_0 \subset \mathbb{K}_1 \subset \dots \subset \mathbb{K}_s$ and
 $\ell_j := [\mathbb{K}_j : \mathbb{K}_{j-1}] = [G_{j-1} : G_j].$

Case I



$\ell_i = [\mathbb{K}_i : \mathbb{K}_{i-1}]$ (= degree of the extension $\mathbb{K}_i/\mathbb{K}_{i-1}$).

$\ell_1 \ell_2 \cdots \ell_s = \#$ complex conjugate branches of C .

Special Curvettes

Let $\pi : (X, D) \rightarrow (\mathbb{C}^2, 0)$ be the minimal G -resolution of C . Let $E_\sigma \subset D$.

- Let $\gamma_\sigma \subset X$ be a smooth and transversal curve to E_σ **defined over the field $\mathbb{K}_\sigma$** . (i.e. the curve γ_σ is invariant under the action of G_σ .)
- Let $C_\sigma = \pi(\gamma_\sigma) = \{f_\sigma = 0\}$ be the blow-down γ_σ , $f_\sigma \in \mathbb{K}[[x, y]]$.

Definition

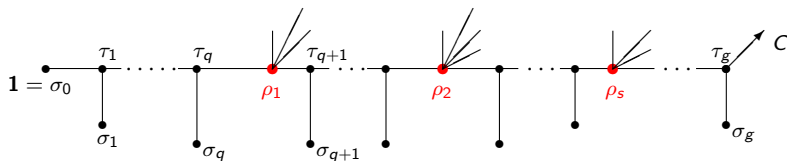
A G_σ -curvette (at the component E_σ) is the union of the curves gC_σ over representatives of the G_σ -classes in G .

A G_σ -curvette can be defined by the equation

$$F_\sigma = \prod_{[g] \in G/G_\sigma} (gf_\sigma) \in \mathbb{K}[[x, y]].$$

- **Let $M_\sigma = \nu(F_\sigma)$** . (Note that $M_\sigma = \sum \nu(gf_\sigma) = \sum m_{g\sigma}$.)

The semigroup Poincaré series



Lemma

Let $\omega \in \check{\Gamma}$ be such that $\rho_d < \omega \leq \rho_{d+1}$ and $\tau_p < \omega \leq \tau_{p+1}$. One has

$$M_\omega = \sum_{i=0}^p k_i M_{\sigma_i} + \sum_{j=1}^d r_j (\ell_j - 1) M_{\rho_j}$$

for some non negative integers k_i ($0 \leq i \leq p$), $k_i < N_i$ for $i \geq 1$, and positive integers r_j ($1 \leq j \leq d$).

Proposition

The integers $M_{\sigma_0}, M_{\sigma_1}, \dots, M_{\sigma_g}$ generate the semigroup $S_\nu^{\mathbb{K}}$ of values of the valuation ν on $\mathbb{K}[[x, y]]$.

Moreover, $(N_i - 1)M_{\sigma_i} \notin \langle M_{\sigma_0}, \dots, M_{\sigma_{i-1}} \rangle$, $N_i M_{\sigma_i} \in \langle M_{\sigma_0}, \dots, M_{\sigma_{i-1}} \rangle$ and $M_{\tau_i} = N_i M_{\sigma_i} < M_{\sigma_{i+1}}$.

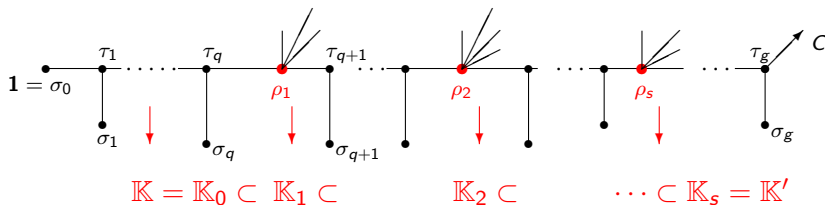
Remark. In particular, $S_\nu^{\mathbb{K}}$ is the semigroup of values of a complex analytic irreducible curve. *why?*

Theorem

One has

$$P_\nu^S(t) = \frac{\prod_{i=1}^g (1 - t^{M_{\tau_i}})}{\prod_{i=0}^g (1 - t^{M_{\sigma_i}})}.$$

The Poincaré series



$\ell_i = [\mathbb{K}_i : \mathbb{K}_{i-1}]$; $\ell_1 \ell_2 \cdots \ell_s = \#$ complex branches of F .

Theorem

$$P_{\nu}^{\mathbb{K}}(t) = P^S(t) \prod_{j=1}^s (1 + t^{M_{\rho_j}} + \cdots + t^{(\ell_j-1)M_{\rho_j}}) = P^S(t) \prod_{j=1}^s \frac{1 - t^{\ell_j M_{\rho_j}}}{1 - t^{M_{\rho_j}}}$$

Remark: The conductor and symmetry

The conductor of the semigroup $S_C^{\mathbb{K}}$ is $c = \sum_{i=1}^g (N_i - 1)M_{\sigma_i} - M_{\sigma_0} + 1$.

Let $\Delta := c + \sum_{j=1}^s (\ell_j - 1)M_{\rho_j}$ and $\ell = \ell_1 \cdots \ell_s = [\mathbb{K}_s : \mathbb{K}]$. One has that

- $a_v = \dim J^{\mathbb{K}}(v)/J^{\mathbb{K}}(v+1) = \ell$ for all $v \geq \Delta$ and $a_{\Delta-1} < \ell$.
- For all $v, w \in \mathbb{Z}$ such that $v + w = \Delta - 1$ one has $a_v + a_w = \ell$.

Following the notations of the Standard Example (Case I), the integral closure $\overline{R}$ of the local ring $R = \mathbb{K}[[x, y]]/(F)$ is isomorphic to $\mathbb{K}'[[u]]$ (u is a uniformizing parameter). One has:

Corollary

The conductor ideal $\mathcal{C}(\overline{R}/R) = \{z \in \overline{R} : z\overline{R} \subset R\}$ is the ideal $u^{\Delta}\overline{R}$.

Note that the ring R is Gorenstein (being plane) and the symmetry above is just the characterization of the Gorenstein condition of the ring R .

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Thank your for your attention